

Quiz 5 - Results



Attempt 1 of 1

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Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

How does Rotary Positional Embedding (RoPE) interact with the attention mechanism's Query and Key vectors?



It rotates Q and K such that their dot product depends only on their relative distance $i-j$.



It applies a fixed, position-dependent 2D rotation ($\sin/\cos$ "phase") to each pair of dimensions in Q and K (leaving V unchanged), so the attention logits incorporate position via these rotated Q/K vectors.



It adds a learnable absolute positional vector to Q and K before the dot product is calculated.



It applies a rotation matrix to the Value V vectors to ensuring the output context is position-aware.

Question 2

1 / 1 point

Which of the following correctly describes the "SwiGLU" FFN variant used in models like LLaMA?

- ☐ It reduces the parameter count of the FFN block by 50% compared to a standard ReLU FFN while maintaining performance.
- ☐ It utilizes a standard ReLU activation function but adds a learnable "Gate" scalar to every neuron.
- ☐ It replaces the standard Feed-Forward Network with a Self-Attention layer to capture gated dependencies.
- ☒ It involves an element-wise product of two linear projections, one of which is activated by the Swish function.

Question 3

1 / 1 point

What is a disadvantage of Pre-Normalization?

- ☐ It makes the model impossible to parallelize.
- ☒ It can be numerically unstable, leading to training spikes.
- ☐ It restricts the use of residual connections.
- ☐ It requires a deeper network to achieve the same accuracy.

Question 4

1 / 1 point

When replacing a standard FFN with a SwiGLU block, why do models (like LLaMA) typically reduce the hidden dimension by a factor of approximately 2/3?

- ☒ To prevent the "Swish" activation function from saturating the gradients during the backward pass.



To maintain a similar total parameter count, as SwiGLU introduces additional weights via its gating mechanism.



To force the model to learn more compact representations by creating a bottleneck in the feed-forward layer.



To align the dimension sizes with the multiple of 256 required by TPU/GPU tensor cores for efficiency.

Question 5

1 / 1 point

Regarding the "Pre-Norm" vs. "Post-Norm" architecture variants, which statement best reflects the trade-offs?



Pre-Norm is mathematically equivalent to Post-Norm during training, but offers faster inference speeds due to layer fusion.



Pre-Norm generally achieves significantly better validation perplexity (accuracy) than Post-Norm on standard benchmarks.



Pre-Norm enables the training for very deep models with Residual Connection but may not offer the exact same numerical optimization properties as Post-Norm.



Post-Norm is prone to "training spikes" and numerical instability, whereas Pre-Norm effectively eliminates the need for a warm-up stage.

Question 6

1 / 1 point

While Layer Normalization has benefits over Batch Normalization for NLP, which following statement is a *disadvantage* of Layer Normalization?



It creates dependencies between samples in the same batch, making it difficult to use with small batch sizes.



It fails to reduce the internal covariate shift (bias) derived from the data.



It requires maintaining running statistics for expectation and variance during inference, which complicates the implementation



It is computationally significantly more expensive than Batch Normalization because it normalizes across the sequence dimension.

Question 7

1 / 1 point

Why is Batch Normalization typically considered less suitable for Transformer-based NLP models compared to Computer Vision models?



The statistics (mean/variance) computed across the batch dimension become unstable with small batch sizes and vary significantly across different sequence positions.



Batch Normalization introduces a high computational overhead because it must compute the covariance matrix for every token in the vocabulary.



Batch Normalization requires fixed-size input images, whereas text sequences have variable lengths that cannot be padded.



Batch Normalization prevents the residual connections from propagating gradients effectively in sequential data.

Question 8

1 / 1 point

What is the primary operational advantage of Grouped-Query Attention (GQA) over Multi-Head Attention (MHA) during inference?



It improves the model's perplexity by allowing multiple heads to share the same Query projection.

- ☐ It allows the model to attend to longer sequences by compressing the Query vectors into groups.
- ✓ ☒ It significantly lowers memory bandwidth usage by reducing the size of the KV cache.
- ☐ It reduces the computational complexity of the attention map from Quadratic

$$O(N^2)$$

to Linear

$$O(N)$$

.

Question 9

1 / 1 point

What is the primary motivation for using Residual Connections (Skip Connections) in deep neural networks?

- ☐ To reduce the total number of parameters in the model.
- ✓ ☒ To enable the training of substantially deeper networks.
- ☐ To perform normalization.
- ☐ To replace the need for Attention.

Question 10

1 / 1 point

What is a unique "Pro" of Sinusoidal Positional Embeddings that distinguishes them from standard learned absolute embeddings?

- ☐ They are computationally cheaper because they do not require an element-wise addition to the input embedding.
- ☐ They allow the model to learn the frequency of the data distribution during the backpropagation phase.
- ☐ They are fully translation invariant, meaning the absolute position of a token never affects the attention score.
- ☒ They theoretically allow the model to extrapolate to sequence lengths longer than those seen during training.

Done